Daniel Kaminski

Email: dgk64@cornell.edu
LinkedIn: linkedin.com/in/daniel-kaminski-447785234
Mobile: +1-508-868-6412

Education

- **Cornell University** — Ithaca NY, USA
  B.S. Electrical & Computer Engineering, GPA: **4.146/4.00** — May 2027
  Courses: *Advanced High Speed & RF ICs, Analog IC Design, Microwave Circuit Design, Microelectronics, Computer Architecture, EM Fields & Waves, Nanosci. & Nanoeng., Signal & System Analysis, Quantum Physics*
  Awards/Honors: *William S. Einwechter Memorial Prize, IEEE Eta Kappa Nu, Eagle Scout*

Work Experience

- **Advanced Micro Devices - Analog Design Engineering Intern (DDR PHY)** — Summer 2026
  Boxborough, MA
  - **Owned two macros** (analog test & clocking calibration) as part of the DDR PHY analog mixed-signal tapeout, improving testbenches, providing collateral, enabling regression, & ensuring specification compliance
  - **Created and tuned voltage regulator model**, ensuring accuracy in key metrics across operating regimes
  - Developed team infrastructure, including SKILL script for V_{TH} flavor symbol fix & job monitoring GUI
- **Analog Devices - Analog Design (MEMS) Internship** — Summer 2025
  Wilmington, MA
  - Implemented several ultra-low power MEMS analog front-end (AFE) charge sense amplifier topologies
  - **Scripted simulation testbenches** to determine key characteristics including bandwidth, noise matching current, output swing, effective transcapacitance, & loopgain (with stability analysis)
  - Derived **design equations** to provide insights on topology feasibility & predict expected performance
- **Analog Devices - Semiconductor Device Engineering Internship** — Summer 2024
  Wilmington, MA
  - **Owned tapeout of R&D GaN power FET test chip**, analyzed & optimized design for key performance metrics, standardized & cellularized layout for scalability
  - Enabled process tool chain, including CalibreDRV DRC & LVS checks, fill, & tape-in scripts
  - **Characterized experimental GaN devices**, analyzed & summarized results using Python
  - Developed internal website to showcase & consolidate all wide bandgap development at ADI

Tapeouts

- **40MS/s "Adiabatic" Flash ADC - Cornell Custom Silicon Systems** — Aug 2025 - Jun 2026
  - Led development of **SCRL adiabatic** (energy recovery) 40MS/s 3-bit Flash ADC in TSMC 180nm
  - Designed nW adiabatic logic cells, 5.2μW energy recovery comparator, 4-phase 16-signal power-clock generator, & ultra-low power reference voltage generator
  - Completed layout (with top and padring), ran through flow (DRC+LVS+fill), and taped out design
- **8-Bit Fully/Truly Differential SAR ADC - Cornell Custom Silicon Systems** — Aug 2024 - Jun 2025
  - **Led team of 5 undergraduate students** to develop an **8-bit 4.44MS/s fully & truly differential SAR ADC** in TSMC 180nm, taped out through MUSE MPW program
  - Designed fully/truly differential toggle mechanism, double tail comparator, charge injection cancellation analog switches, CDAC, & digital control logic; completed layout and padring design

Project & Research Experience

- **Molnar Group Research Assistant** — Sep 2025 - Present
  - Assisted in design of high-speed 8-phase passive mixer in 22nm FDSOI process
- **Microelectronics (ECE3150) & Analog IC Design (ECE4530) Teaching Assistant** — Jan 2026 - Present
- **2.5GB/s Plesiochronous SERDES in GPDK45** — Oct 2025 - Dec 2025
  - Designed decision feedback equalization & continuous time linear equalization circuitry in GPDK45 for analog design course final project, performed layout, & created technical report
- **Cornell Custom Silicon Systems (C2S2) - Team Lead** — Oct 2023 - May 2026
  - Lead of team with over 55 members, previously analog subteam lead
  - Organized events and team meetings, managed >$100k of sponsor funds (AMD, Sandia National Labs, & NORDTECH), directed team recruitment, and set team direction
- **Jena-Xing Group Research Assistant** — Aug 2024 - Mar 2025
  - Characterized DrGaN circuits, determining key performance metrics ($f_t, R_{dson} \times Area$, & Q_{sw})

Technical Skills

- **Skills:** Analog/Mixed-Signal IC Design (Data Converters, Wireline, RF), Microwave Design, Embedded Prog.
- **Tools:** Cadence Virtuoso, Spectre, Calibre DRV, Intel Quartus, Xilinx Vivado, XSchem, Magic VLSI, Git, Linux
- **Programming Languages:** SKILL, Verilog, Bash, Python, C/C++